



Detection of Dangerous Object on Baggage Inspection using Deep Learning in X-ray images

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0-Abstract

In the field of security, an increasingly non-invasive method is the analysis of elements using X-rays. Particularly at airports, this is the common protocol when it comes to detecting possible dangerous objects in passenger baggage. Unfortunately, such human inspection only can to detect around 80% – 90%, under optimal operator conditions. Due to the use of automatic detectors for this type of objects can be of great help combating these risks. This work propose a methodology centering on implement modern algorithm of deep learning to dangerous object detection on baggages, using X-ray images. For this purpose, the methodology use simulations algorithms to replicate real baggages, and have enough images to train correctly the deep learning algorithms as YOLO, SSD and RetinaNet. When employ this methodology, was obtain miscellaneous object detection for 4 dangerous object: knife, razor, gun and shuriken. These detectors are evaluate on independent set with the usual metric mAP and the processing time per image. The best result was 80% of the YOLOv3 model, more precisely: gun 96.3%, knife 76.2%, razor 86.9% and shurikens 93.7%. This research prove efficiency to the model of object detection and it possible use on real security environment. Moreover, this research expose a methodology to create scalable train images of dangerous object, it allow train elaborate models without a complex data base of images.

2-Images Simulation: Strategy of Superimpose

The central idea of the method is to superimpose the dangerous objects in the luggage, thus obtaining a luggage with dangerous elements inside, as presented in (D. Mery and Katsaggelos, 2017). This is based on the use of the law of absorption that characterizes the distribution of X-rays through matter. We know that a gray scale X-ray image is followed by a linear model, defined by Equation 1 where A and B are constant parameters of the capture model and φ is the intensity registered by the sensor.

$$I = A \cdot \varphi + B \quad (1)$$

Using this model, we can define the image of the dangerous object (I_p) and the image of luggage (I_b) as:

$$I_p = A \cdot \varphi_p + B \quad I_b = A \cdot \varphi_b + B \quad (2)$$

where

$$\varphi_p = \varphi_0 e^{-u_p d_p} \quad \varphi_b = \varphi_0 e^{-u_b d_b}. \quad (3)$$

So, to get the combined image I_t modeled as:

$$I_t = A \cdot \varphi_t + B = A \cdot \varphi_0 e^{-(u_b d_b + u_p d_p)} + B \quad (4)$$

Combining the Equations 2, 3, 4 and defining $C = A\varphi_0$ we get that:

$$I_t = C \cdot J_p \cdot J_b + B \quad \text{where} \quad J_p = \frac{I_p - B}{C} \quad J_b = \frac{I_b - B}{C}. \quad (5)$$

An example of the method is shown in Figure 1, it can be seen that it is practically impossible to discriminate that the result is not from a real image.



Figure 1: Strategy of Superimpose.

4-Conclusion

This research has as contribution to be an advance in the study of detection of threat objects in X-rays. Through the proposed methodology, emphasis is placed on the ease in creating simulated images and their almost complete independence from the test images. In this way, a solid alternative is generated for the training of the new deep learning models, without the need to have access to more complex databases and difficult public concession. Together with the proposal of a clear training structure that can be replicated, and therefore compared to future research that wishes to corroborate its methodology.

In terms of results, through the YOLOv.3 Redmon and Farhadi, 2018 model, yields around 90% (wmAP) were achieved, while the second most successful model, RetinaNet Lin, Goyal, Girshick, He, and Dollár, 2017, achieves an average of 80% (wmAP), in the different folders of test images that contained similar threat objects with which you trained. Especially the Gun and Shuriken classes obtain results that exceed human operators (between 80% and 90%)

1-Security Inspection

Security in different places, both public and private, has been of utmost importance over the centuries with the aim of preventing crimes of different kinds, spread of pests or diseases, among other reasons. This security has been focused mainly on the visual detection of dangerous elements, be it weapons, objects, animals or organic elements not allowed. In the particular case of airports, security has been extremely important for the authorities, so different technologies have been implemented for the benefit of security. This is the case of the non-invasive analysis of passenger baggage through the use of X-rays, which allows objects to be analyzed quickly avoiding the need for direct contact with the passenger's baggage. Unfortunately, X-ray inspection processes are complicated due to various components. One of these factor is human performance, the literature indicates that these operators only reach a yield between 80% and 90%. It is for this reason that developing different techniques for the automatic detection of objects by means of images obtained with X-rays, can become a great help to assist the work of security operators.

3-Experiment and Result

This work uses the public database GDXray (D. Mery et al., 2015) from which images corresponding to four types of isolated dangerous objects were used, these objects are classified under the classes: Razor, Guns, Knives and Shurikens. For the creation of the **training set** the following folders are used for each class:

- Gun: B0049 (200 images).
- Knife: B0076 (576 images).
- Razor: B0051 (100 images).
- Shuriken: B0052 (144 images).
- Other : B0082 (600 images).
- Baggage: BX_100 (48 images)

The experiment consists of simulating 200 samples per image in our backpacks set BX100. In each of these simulations, elements are superimposed on randomly chosen positions on both dangerous and non-dangerous objects. Therefore, there are 9800 training images with their respective annotations of the position of significant objects. The validation set was defined as a subset of the simulated training data, specifically as a tenth of this.

Model \ AP	Gun	Knife	Razor	Shuriken	mAP	wmAP	time
SDD300	0.901	0.011	0.696	0.904	0.630	0.776	0.036
RetinaNet	0.981	0.005	0.819	0.956	0.690	0.857	0.131
Yolo2	0.780	0.054	0.607	0.880	0.580	0.700	0.056
Yolo3	0.963	0.762	0.869	0.937	0.883	0.910	0.182



Figure 2: Example detection gun.